

Executive Summary

Meeting of ITU-T SG17 'Security', virtual, 10-20 May 2022

Hot topics

- 5G security
- Countering spam
- QKDN
- Intelligent transport system security
- DLT based security services
- IoT security
- Security multi-party computation

1 Meeting Output (meeting statistics see Annex E below)

- SG17 Working Party structure: no change from last Study Period.
- Default approval procedure for SG17 Questions: no change from last Study Period.
- SG17 Rapporteurship: same as last Study Period, adding one new Q13/17 associate Rapporteur
- Output standards (12, see Annex A):
 - TAP approval (3): Details are in Annex A a).
 - TAP determined (3): 3 new Recommendations. Details are in Annex A b).
 - AAP consented (3): 3 new Recommendations for AAP Last Call. Details are in Annex A c).
 - Agreed (3): 3 new Technical Reports. Details are in Annex A d).
- New work item established (1517, see Annex C).
- Work items discontinued (1): Detail is in Annex D.
- New Registry authority for Republic of Korea: {joint-iso-itu-t(2) country(16) kr(410)}
- Workshop: [1st ITU-T X.509 Day Event](#) was held successfully on 9 May 2022.
- SG17 JCAs, Correspondence Group/task force
 - JCA-IdM: continued
 - 3 Correspondence Groups
 - CG-secapa (Correspondence Group on Security Capability and Architecture): new
 - CG-SG17-meeting (Correspondence Group on effective SG17 meeting): continued
 - CG-FIDO (of Q10/17): new

2 Next SG17 meetings

2.1 2nd SG17 meeting: Geneva, Tuesday 23 August – Friday 2 September 2022 (9 working days, physical meeting with remote participation (as WTSA-20))

- ITU workshop on security for 5G and beyond, Monday 22 August 2022, 09:30-17:30
- Open and extended management team meeting on Monday 22 August 2022, 19:00-21:00.

- 30th JCA-IdM meeting on Friday 26 August 2022, 14:30-16:00
- 20 Candidate texts planned for action, details see Annex B.

2.2 3rd SG17 meeting: Geneva, Tuesday 21 February - Friday 3 March 2023 (9 working days)

- Open and extended management team meeting on Monday 20 February 2023, 19:00-21:00.

2.3 Interim RGMs

6 Questions plan to hold 6 RGMs before next SG17 meeting

#	Q	Date	Place/Host	Subject/objective
1.	2/17	30 June 2022	e-meeting	• prepare texts for action in next SG17 meeting: X.5Gsec-ecs and X.5G-ssl
2.	3/17	June-July 2022	e-meeting	• Discuss survey for CDC in Africa
3.	10/17	July 2022	e-meeting	• progress all the work of Q10/17
4.	11/17	27 June – 01 July 2022	e-meeting	• Joint meeting with ISO/IEC/JTC 1/SC6, agenda in TD216
5.	13/17	7-8 June 2022	e-meeting	• prepare final texts for X.ipscv, X.edr-sec, X.eivn-sec, X.srcl, X.fistiscv • address all work items
6.	14/17	12-13 July 2022	e-meeting	• work on the action items: X.srscm-dlt, X.sadsm • review of deliverables from other study groups, focus groups and other SDOs

Annex A

Actions taken on Recommendations, and other texts at SG17 closing plenary on 20 May 2022

a) TAP Recommendations approved (WTSA-20 Resolution 1)

#	Q	Acronym	Title	New / Revised	Editor(s)	Location of text	Equivalent e.g., ISO/IEC	Start of work	Timing
1.	2/17	X.1812 (X.5Gsec-t)	Security framework based on trust relationship for IMT-2020 ecosystem	New	Minpeng Qi, Li Su, Junzhi Yan, HeungYoul Youm	TD128		2018-09	2021-09
2.	4/17	X.1246Amd.1	Technologies involved in countering voice spam in telecommunication organizations	New	Dmitry Cherkosov, Yanbin Zhang	TD322		2019-09	2021-09
3.	4/17	X.1247Amd.1	Technical framework for countering mobile messaging spam	New	Dmitry Cherkosov, Yanbin Zhang	TD323		2019-09	2021-09

b) TAP Recommendations determined (WTSA-20 Resolution 1)

Q	Acronym	Title	New / Revised	Editor(s)	Location of Text	Equivalent e.g., ISO/IEC	Start of work	Timing

2/17	X.1813 (X.5G-vs)	Security requirements for operation of vertical services supporting ultra-reliable and low latency communication (URLLC) in IMT-2020 private network	New	Jae Eon Oh, Seongki Shin, Heung Youl Youm	TD300		2020-09	2022-05
2/17	X.1814 (X.5Gsec-guide)	Security guideline for 5G communication system	New	Mee Yeon Kim, Keundug Park, Heung Youl Youm	TD264		2019-01	2022-05
6/17	X.1352 (X.iotsec-4)	Security requirements for IoT device and gateway	New	Jiho Bang, Wonsuk Chung, Sanggeol Lee, Hosoek Ryu	TD269		2018-09	2022-05

c) AAP Recommendations consented (Recommendation ITU-T A.8)

Q ⁽¹⁾	Acronym	Title	New / Revised	Editor(s)	Location of Text	Equivalent e.g., ISO/IEC	Start of work	Timing
13/17	X.1379 (X.rsu-sec)	Security requirements for road-side unit in intelligent transportation system	New	Qiguang Fan, Ye Tian, Junzhi Yan	TD271		2019-09	2022-05

14/17	X.1409 (X.ss-dlt)	Security services based on distributed ledger technology	New	Yue Chen, Zhaoji Lin, Min Shu, Ke Wang, Kai Wei, Junjie Xia, Junzhi Yan	TD222		2017-05	2022-05
15/17	X.1715 (X.sec-QKDN-intrq)	Security requirements and measures for integration of QKDN and secure storage network	New	Kaoru Kenyoshi	TD325		2020-09	2022-05

d) Non-normative texts (Technical Report, Supplement, Implementers' Guide, etc) agreed

Q	Acronym	Title	New / Revised	Editor(s)	Location of text	Equivalent e.g., ISO/IEC	Start of work	Timing
2/17	XSTP-5Gsec-RM	Technical Report: 5G Security Standardization Roadmap	New	Yutaka Miyake, Junzhi Yan, Gunhee Lee, Stiepan Kovac	TD312		2021-09	2022-05
15/17	TR.sec-ai	Technical Report: Guidelines for security management of using artificial intelligence technology	New	Linlin Zhang, Liang Wei, Huiyun Jing, Junjie Xia	TD232			2022-05

15/17	TR.hyb-qkd	Technical Report: Overview of hybrid approaches for key exchange with QKD	New	Dong-Hi SIM	TD301			2022-05
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Annex B

Recommendations planned for action in SG17 Aug/Sep 2022 meeting

a) TAP Recommendations planned for TAP approval (WTSA-20 Resolution 1)

Q	Acronym	Title	New / Revised	Editor(s)	Location of Text	Equivalent e.g., ISO/IEC	Start of work	Timing
2/17	X.1813 (X.5G-vs)	Security requirements for operation of vertical services supporting ultra-reliable and low latency communication (URLLC) in IMT-2020 private network	New	Jae Eon Oh, Seongki Shin, Heung Youl Youm	R7		2020-09	2022-05
2/17	X.1814 (X.5Gsec-guide)	Security guideline for 5G communication system	New	Mee Yeon Kim, Keundug Park, Heung Youl Youm	R8		2019-01	2022-05
6/17	X.1352 (X.10sec-4)	Security requirements for IoT device and gateway	New	Jiho Bang, Wonsuk Chung, Sanggeol Lee, Hosoek Ryu	R9		2018-09	2022-05

b) TAP Recommendations planned for TAP determination (WTSA-20 Resolution 1)

#	Q ⁽¹⁾	Acronym	Title	New / Revised	Editor(s)	Location of Text	Start of work	Timing
1.	2/17	X.5Gsec-ecs	Security guidelines for 5G edge computing services	New	Feng Gao, Jae Hoon Nah, Junjie Xia, Bo Yu, Xiaojun Zhuang	TD311	2019-01	2022-09
2.	2/17	X.5Gsec-ssl	Guidelines for classifying security capabilities in 5G network slice	New	Zhiyuan Hu, Li Su, Ke Wang, Bo Yang	TD245	2020-09	2022-09
3.	6/17	X.sc-iot	Security Controls for Internet of Things (IoT) system	New	Koji Nakao, Liu Lijun	TD4091	2018-09	2022-09
4.	6/17	X.ztd-iot	Security methodology for zero-touch deployment in massive IoT based on blockchain	New	Xin Kang, Haiguang Wang, Weidong Wang	TD243	2020-09	2022-09
5.	7/17	X.sles	Security measures for location enabled smart office service	New	Hao Dong, Lijun Liu, Jae Hoon Nah, Wenxin Wang	TD295	2019-09	2022-09
6.	8/17, (14/17)	X.BaaS-sec	Guidelines on blockchain as a service (BaaS) security	New	Nan Meng, Kyeong Hee Oh	TD231	2019-09	2022-09
7.	8/17	X.sgdc	Security guidelines for distributed cloud	New	Jinfeng Kou, Mark McFadden, Ye Tao, Laifu Wang, Lei Xu	TD252	2019-01	2022-09
8.	13/17	X.srzd	Security requirements for	New	Yaping Sun,	TD285	2018-03	2022-09

			categorized data in V2X communication		Huirong Tian, Nan Meng, Takamasa Isohara			
9.	13/17	X.edr-sec	Security guidelines for cloud-based event data recorders in automotive environment	New	Sang-Woo Lee, Seungwook Park	TD281	2018-09	2022-09
10.	13/17	X.eivn-sec	Security guidelines for Ethernet-based In-Vehicle network	New	Sang-Woo Lee, You-Sik Lee	TD279	2018-09	2022-09
11.	13/17	X.fstiscv	Guidelines for sharing security threat information on connected vehicles	New	Min Shu, Wenlei Wang, Yunwei Zhao, Xiaochun Yun	TD282	2018-09	2022-09
12.	14/17	X.sa-dsm	Security architecture of data sharing management based on the distributed ledger technology	New	Min Shu, Yunei Zhao, Feng Gao, Zhiyuan Hu	TD293		2022-09

Notes:

(1) In case of joint Question activity, the lead Question is given without parentheses and other Questions are shown in parentheses; such entries are only shown in the table against the lead Question.

c) AAP Recommendations planned for AAP consent (Recommendation ITU-T A.8)

Q ⁽¹⁾	Acronym	Title	New / Revised	Editor(s)	Location of Text	Start of work	Timing
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1/17	X.arch-design	Design principles and best practices for security architectures	New	Arnaud Taddei	TD3386	2019-09	2022-09
13/17	X.ipscv	Guidelines for an intrusion prevention system for connected vehicles	New	Seonghoon Jeong, HuyKang Kim, Sang-Woo Lee, Seungwook Park	TD277	2019-09	2022-09
14/17	X.srscm-dlt	Security Requirements for Smart Contract Management based on distributed ledger technology	New	Tietao Guo, Li Li, Kyeong Hee Oh, Min Shu, Yang Wu	TD236	2020-09	2022-09

d) Non-normative texts (Technical Report, Supplement, Implementers' Guide, etc) planned for agreement

Q	Acronym	Title	New / Revised	Editor(s)	Location of text	Start of work	Timing
4/17	X.Sup-cs-ml	Supplement to X.1231: Countering spam based on machine learning	New	Elizabeth Essl, Wei Liu, Weidong Wang, Lei Xu, Yanbin Zhang	TD239	2019-08	2022-09
7/17	TR.cta	Technical Report: Use cases for contact tracing technologies to prevent spread of infectious diseases	New	Mee Yeon Kim, Heung Youl Youm	TD294	2020-09	2022-09

Annex C

New work items

The following new work items were agreed to be added to the SG17 Work Programme:

#	Q	NWI	Approval	TD	Title	C
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1.	2/17	TR.cpn-col-sec	Agreement	TD248	Technical Report: Security considerations of collaboration of multiple computing power networks	C009
2.	2/17	TR.5Gsec-bsf	Agreement	TD255	Technical Report: Guidelines of built-in security framework for telecommunications network	C056
3.	2/17	X.5Gsec-srocv	TAP	TD268	Security Requirements for the Operation of IMT-2020 Core Network to Support Vertical Services	C073
4.	4/17	X.sr-ctea	AAP	TD244	Security requirements and countermeasures for targeted email attacks	C021
5.	4/17	X.spmoh	AAP	TD241	Security framework for storage protection against malware attacks on hosts	C042
6.	4/17	X.sgc_rcs	TAP	TD286	Guidelines for countering spam over rich communication service (RCS) messaging	C087
7.	6/17	TR.ba-iot	Agreement	TD321	Technical Report: Broadcast authentication scheme for IoT system	C074
8.	7/17	X.suppl.uc-dcc	Agreement	TD266	Supplement to X.1152 on use cases for digital COVID-19 certificates	C031
9.	7/17	X.smdtf	AAP	TD258	Security measures for digital twin federation in smart cities and communities	C061
10.	7/17	X.srmcp	AAP	TD259	Security requirements for monitoring physical city assets	C062
11.	7/17	X.tc-ifd	TAP	TD253	Technical capabilities of interactive fraud detection	C093
12.	8/17	X.soar-cc	TAP	TD272	Framework of security orchestration, automation and response for cloud computing	C067
13.	8/17	X.gdsml	TAP	TD229	Guidelines for data security using machine learning in big data infrastructure	C088
14.	10/17	X.ctap21	AAP	TD214	FIDO Client to Authenticator Protocol 2.1	TD214
15.	10/17	X.uaf12	AAP	TD214	FIDO Universal Authentication Framework Protocol Specification 1.2	TD214
16.	15/17	X.seca-def	AAP	TD288	Security capabilities definitions	C013
17.	15/17	TR.smpa	Agreement	TD319	Technical Report: Security middle platform architecture	C099

Annex D

Work items discontinued

Question	Acronym	Title
10/17	X.tec-idms	Management and protection techniques for user data protection in distributed identity systems

Annex E

SG17 meeting Statistics

Participants ([TD2R1](#))

	participants	countries	Member States	Sector Members	SG17 Associates	Academia	Invited Experts
Announced	276	39	36	32	1	6	6
Final	281	39	35	32	1	6	6

Meeting input and organization

Table of SG17 statistics of this and some past meetings

	2022-05*	2021-09*	2021-04*	2020-09*	2020-03*	2019-09	2019-01	2018-09	2018-03	2017-09	2017-03	2016-09	2016-03	2015-09	2015-04
C	101	98	104	110	121	151	118	144	113	106	78	81	66	74	80
LS/i	72	46	57	58	49	67	58	47	40	45	43	38	47	56	60
LS/o	20	26	17	28	29	40	39	37	38	46	38	23	32	42	42
TD	331	336	307	434	438	508	380	420	395	426	368	391	418	371	386

Note * - fully virtual meeting

Contribution: 101 – stable, DDP: 97%.

- APT 93 (93%) [China 44 + Korea 39 + Japan 9 + Singapore 1]
- Americas 4 (4%) [US 4]
- RCC 2 (2%) [Russia 2]
- EUR 1 (1%) [UK 1]
- AFR (0), ARAB (0), LAM (0)

Liaison Statement: matrix in [TD23](#)

- incoming 72 - increased due to longer interval since last SG17 meeting
- Outgoing 20 - stable

TDs: 331 - stable